Dynamics and relativity

Oliver West

Contents

Lecture 1 – 22/01/26

§1 The structure of the Newtonian universe

- Three dimensional space endowed with a *Cartesian reference frame* – an origin and some axes.
- Time can be labelled in an arbitrary reference frame, by a real number t.

- A *point particle* is an idealised object that is completely determined by its position at a given point in time, $\mathbf{x}(t)$. Its velocity

  $$\mathbf{v} = \frac{d\mathbf{x}}{dt} = \left(\frac{dx_1}{dt}, \frac{dx_2}{dt}, \frac{dx_3}{dt}\right) = \dot{\mathbf{x}}$$

  is tangent to the trajectory. The acceleration

  $$\mathbf{a} = \frac{d^2\mathbf{x}}{dt^2} = \left(\frac{d^2x_1}{dt^2}, \frac{d^2x_2}{dt^2}, \frac{d^2x_3}{dt^2}\right) = \ddot{\mathbf{x}} = \dot{\mathbf{v}}$$

The above is *not* sufficient to write down Newton's equations. Consider, for example, a free particle not experiencing any forces. The position of this particle is $\mathbf{x}(t)$, but which frame should we use?

§1.1 Law of inertia

Proposition 1.1 (Law of inertia)

There exist *inertial frames* in which a free particle has a constant velocity.

This is an improved version of Newton's first law. Indeed, it is a true statement, but not an obvious one.

§1.2 Galilean relativity principle

Definition 1.2. A *Galilean transformation* has the form

$$\mathbf{x}' = R\mathbf{x} + \mathbf{k} + \mathbf{w}t$$

where $R \in O(3)$, $\mathbf{k}$ is a translation vector, and $\mathbf{w}$ is a *boost*. In particular, $\ddot{\mathbf{x}}' = \ddot{\mathbf{x}}$.

Proposition 1.3 (Galilean relativity principle)

A frame related to an inertial frame by a *Galilean transformation* is also an inertial frame, and all laws of physics are the same in both frames.

Example of a boost: dropping a ball on a boat moving at constant velocity.

Galilean invariance also restricts the forces that are possible (see example sheet). For example, there are no 'special points' in the universe (or special directions, times, velocities, etc) – they are all relative. This is *not* the case for acceleration – an object accelerating in one inertial frame is accelerating with the same magnitude in all inertial frames.

Galilean transformations form the *Galilean group*, often supplemented by shifts in time, under which the laws of physics are also invariant. In particular, this implies that all inertial frames have the same time, called absolute time.

§2 Forces

Once there is more than one particle in the universe, there will be interactions between them. In Newtonian physics, these are described by forces.

§2.1 Newton's second law

Theorem 2.1 (Newton's second law)
In an inertial frame,

$$\dot{\mathbf{p}} = \mathbf{F}$$

where the *momentum* $\mathbf{p}$ is $m\mathbf{v}$, where m is the *inertial mass*, mass for short.

The mass is an additional property for point particles that we take to be constant in time unless otherwise stated.

The force $\mathbf{F}$ depends on the ongoing interactions, but it can only depend (axiomatically) on $\mathbf{x}$ and $\mathbf{v}$. This implies that Newton's second law is a second-order differential equation, and that, given $\mathbf{x}$ and $\mathbf{v}$ at $t = 0$ for all particles, Newton's equations will uniquely determine $\mathbf{x}(t)$ for all future times.

We should note that Newtonian mechanics has been superseded by quantum mechanics (for small things) and general relativity (for fast things).

§2.2 Conservative forces

Definition 2.2. A force $\mathbf{F}$ is called *conservative* if it can be written

$$\mathbf{F} = -\nabla V$$

for some *potential* scalar field V.

For example, *gravitational potential energy* of a particle of mass m at $\mathbf{x}$ due to a particle of mass M at $\mathbf{x}_0$ is

$$V = -\frac{GMm}{|\mathbf{x} - \mathbf{x}_0|}$$

where G is a constant.

Now we can get the force by taking the gradient. First,

$$\partial_i \left(|\mathbf{x} - \mathbf{x}_0|^2\right) = 2|\mathbf{x} - \mathbf{x}_0|\partial_i|\mathbf{x} - \mathbf{x}_0|$$

But also,

$$\begin{aligned}
\partial_i \left(|\mathbf{x} - \mathbf{x}_0|^2\right) &= \partial_i \left((\mathbf{x} - \mathbf{x}_0)_j(\mathbf{x} - \mathbf{x}_0)_j\right) \\
&= 2(\mathbf{x} - \mathbf{x}_0)_j\partial_i(x_j - (x_0)_j) \\
&= 2(\mathbf{x} - \mathbf{x}_0)_j\delta_{ij} \\
&= 2(\mathbf{x} - \mathbf{x}_0)_i
\end{aligned}$$

and so we have

$$\begin{aligned}
\partial_i|\mathbf{x} - \mathbf{x}_0| &= \frac{(\mathbf{x} - \mathbf{x}_0)_i}{|\mathbf{x} - \mathbf{x}_0|} \\
\implies \boxed{\nabla|\mathbf{x} - \mathbf{x}_0| = \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|}}
\end{aligned}$$

Hence

$$\mathbf{F} = -\nabla V = -GMm\frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|^3}$$

Let $\mathbf{r} = \mathbf{x} - \mathbf{x}_0$, then

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ is a unit vector. Thus we have the inverse square law of gravity.

> **Remark.** Sometimes we write $V = m\Phi$ where $\Phi = -\frac{GM}{|\mathbf{x}-\mathbf{x}_0|}$ is called the *gravitational potential.*

Near the surface of the Earth, take $\mathbf{x}_0 = \mathbf{0}$, and $|\mathbf{x}| = R + z$, where R is the radius of the Earth and $z \ll R$. Then

$$\begin{aligned}\Phi(R+z) &= -\frac{GM}{R+z} \\ &= -\frac{GM}{R}\left[1 - \frac{z}{R} + \frac{z^2}{R^2} + \cdots\right] \\ &\approx \text{constant} + \underbrace{\frac{GM}{R^2}}_{g} z\end{aligned}$$

where $g \approx 9.8\text{ms}^{-2}$. Since the constant will drop out when taking the gradient, we ignore it, so we have

$$\begin{aligned}&\Phi \approx gz \\ \implies\ &\mathbf{F} = -mg\hat{\mathbf{z}}\end{aligned}$$

This force leads to the simplest non-trivial example of motion due to a force. By Newton's second law, we have

$$m\ddot{\mathbf{x}} = m\mathbf{g}$$

where $\mathbf{g} = (0, 0, -g)$. We need only consider the z component, which gives

$$\begin{aligned}&m\ddot{z} = -mg \\ \implies\ &\dot{z} = v_0 - gt \\ \implies\ &z = z_0 + v_0 t - \frac{1}{2}gt^2\end{aligned}$$

for an initial height z_0 and an initial velocity v_0.

§2.3 Energy

Proposition 2.3

Conservative forces have a *conserved energy*

$$\begin{aligned}E &= \frac{1}{2}m\dot{\mathbf{x}}\cdot\dot{\mathbf{x}} + V(\mathbf{x}) \\ &= \frac{1}{2}m\dot{x}_i\dot{x}_i + V(\mathbf{x})\end{aligned}$$

Proof. We have

$$\begin{aligned}\frac{dE}{dt} &= m\dot{x}_i\ddot{x}_i + \frac{\partial V}{\partial x_i}\dot{x}_i \\ &= m\dot{x}_i\left(m\ddot{x}_i + \frac{\partial V}{\partial x_i}\right) \\ &= 0 && (m\ddot{\mathbf{x}} = -\nabla V)\end{aligned}$$

∎

Example 2.4

Suppose we throw an object into space and want it to never fall back down. The minimal velocity required is called the escape velocity.

When the object is thrown, its energy is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

To not fall back, the object must reach $|\mathbf{x}| \to \infty$ without the velocity going to zero. We have

$$E_\infty = \frac{1}{2}mv_\infty^2 - 0$$

By conservation of energy, $E = E_\infty$, and so to make $v_\infty > 0$ we must have

$$\begin{aligned}\frac{1}{2}mv^2 &> \frac{GMm}{R} \\ \implies v^2 &> \frac{2GM}{R} \\ \implies v &> \sqrt{\frac{2GM}{R}}\end{aligned}$$

which is approximately 10kms^{-1} for Earth.

Note: the mass cancels in v_{escape} because the *gravitational mass* (from the inverse square law) is equal to the *inertial mass* (that appears in Newton's second).

Proposition 2.5 (Principle of equivalence)

Objects have the same gravitational and inertial masses.

Remark. This is not known and there is no reason why we would expect this to be true – whether there may be tiny deviations between the masses is under investigation.

It's useful to write energy as $E = T + V$, where T is the *kinetic energy* $\frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}}$.

Definition 2.6. The *work done* by a force as a particle moves along a trajectory C is given by

$$W = \int_C \mathbf{F} \cdot d\mathbf{x}$$

Proposition 2.7

For a conservative force, the work done along a trajectory only depends on the endpoints of the trajectory, not on the path itself.

Proof 1. We have

$$
\begin{aligned}
W = \int_C \mathbf{F}\cdot d\mathbf{x} = \int_{t_1}^{t_2} \mathbf{F}\cdot\frac{d\mathbf{x}}{dt}\,dt
&= m\int_{t_1}^{t_2} \ddot{\mathbf{x}}\cdot\dot{\mathbf{x}}\,dt \\
&= \frac{1}{2}m\int_{t_1}^{t_2} \frac{d}{dt}(\dot{\mathbf{x}}\cdot\dot{\mathbf{x}})\,dt \\
&= T(t_2) - T(t_1) \\
&= V(t_1) - V(t_2) \qquad\qquad (T_2 + V_2 = T_1 + V_1) \\
&= V(\mathbf{x}(t_1)) - V(\mathbf{x}(t_2)) \\
&= V(\mathbf{x}_1) - V(\mathbf{x}_2)
\end{aligned}
$$

from which it's obvious that the work done only depends on the endpoints. ■

Proof 2. We have

$$
\begin{aligned}
W = \int_C \mathbf{F}\cdot d\mathbf{x} &= -\int_C \nabla V\cdot d\mathbf{x} \\
&= V(x_1) - V(x_2)
\end{aligned}
$$

■

§2.4 Electromagnetic forces

Forces that depend on velocity typically don't have a conserved energy (e.g. friction), but the Lorentz force is an exception.

Electromagnetic fields $\mathbf{E}$ and $\mathbf{B}$ exert the following force on a particle with *charge* q:

$$\mathbf{F} = q\left[\mathbf{E}(\mathbf{x}) + \dot{\mathbf{x}}\times\mathbf{B}(\mathbf{x})\right]$$

We will restrict to static electromagnetic fields, in which case we have

$$\mathbf{E} = -\nabla\Phi$$

for some electric potential Φ. We claim the conserved energy is $E = \frac{1}{2}m\dot{\mathbf{x}}^2 + q\Phi(\mathbf{x})$. We have

$$
\begin{aligned}
\frac{dE}{dt} &= m\ddot{\mathbf{x}}\cdot\dot{\mathbf{x}} + q\nabla\Phi\cdot\dot{\mathbf{x}} \\
&= (\mathbf{F} + q\nabla\Phi)\cdot\dot{\mathbf{x}} \\
&= q(\dot{\mathbf{x}}\times\mathbf{B})\cdot\dot{\mathbf{x}} \\
&= 0
\end{aligned}
$$

The velocity-dependent force is orthogonal to the trajectory of the particle, so it does no work on the particle and energy is still conserved.

Electric forces are similar to gravitational ones: the potential Φ at $\mathbf{x}$ due to another particle of charge Q at $\mathbf{x}_0$ is

$$\Phi = \frac{Q}{4\pi\varepsilon_0} \cdot \frac{1}{|\mathbf{x} - \mathbf{x}_0|}$$

where ε_0 is the *permittivity of free space*. Just like with gravity, this leads to an inverse square law called the *Coulomb force.*

An important distinction from gravity is that charges can be positive or negative, but mass is always positive.

Magnetic forces are different. A charged particle in a magnetic field obeys

$$m\ddot{\mathbf{x}} = q\dot{\mathbf{x}} \times \mathbf{B}$$

This is a vector differential equation, which we can solve by writing out in Cartesian coordinates (we will also take $\mathbf{B}$ constant and in the z-direction, $\mathbf{B} = (0, 0, B) = B\hat{\mathbf{z}}$). The equations become

$$\begin{align} m\ddot{x} &= q\dot{y}B \tag{1} \\ m\ddot{y} &= -q\dot{x}B \tag{2} \\ m\ddot{z} &= 0 \tag{3} \end{align}$$

The first thing we can observe is that $z = z_0 + v_z t$, that is, the particle moves with constant velocity in the z-direction. Now, differentiating (1) and subbing into (2), we have

$$m\dddot{x} = q\ddot{y}B = -\frac{q^2\dot{x}B^2}{m}$$

and this is a second-order equation for $\dot{x}$ – we have

$$\dot{x} = \tilde{A}\sin(\omega t + \phi)$$

where $\omega = \frac{qB}{m}$ is the *cyclotron frequency.* Hence

$$\boxed{x = x_0 + A\cos(\omega t + \phi)}$$

Plugging into (1), we have

$$\begin{gathered} qB\dot{y} = -mA\omega^2\cos(\omega t + \phi) \\ \implies \boxed{y = y_0 - A\sin(\omega t + \phi)} \end{gathered}$$

Hence, the particle is moving in a circle in the x-y plane while moving with a constant velocity in the z-direction.

DIAGRAM.

Alternatively, let $\xi = x + iy$, and write $(1) + i(2)$. We have

$$m\ddot{\xi} = -iqB\dot{\xi}$$

which we can solve nice and easily: $\xi = c_1 e^{-i\omega t} + c_2$. Let's put

$$\begin{aligned} c_1 &= Ae^{-i\phi} \\ c_2 &= x_0 + iy_0 \end{aligned}$$

Plugging in and taking real and imaginary parts, we recover the exact same solution as before.

> **Remark.** There is something 'complex' underlying cyclotron motion – c.f. quantum Hall effect.

§2.5 Motion in one dimension

Often, problems involving higher dimensions can be reduced to one-dimensional motion. In such a case, we have

$$m\ddot{x} = F_x$$

If F_x is independent of velocity, it can always be written as the gradient of a potential by setting

$$V(x) = -\int_{x_0}^{x} F_x(x')dx'$$

for an arbitrary reference point x_0. So the energy

$$E = \frac{1}{2}m\dot{x}^2 + V(x)$$

is always conserved. Then, putting $E =$ constant, we get a first-order differential equation for $x(t)$ which is easily integrated: we have

$$\begin{aligned} \dot{x} &= \pm\sqrt{\frac{2}{m}\left(E - V(x)\right)} \\ \implies t - t_0 &= \pm\int_{x_0}^{x} \frac{1}{\sqrt{\frac{2}{m}\left(E - V(x')\right)}}\,dx' \end{aligned}$$

and so solving Newton's equations in one dimension essentially just amounts to doing this integral.

Lecture 4 – 29/01/26

From

$$E = \frac{1}{2}\dot{x}\cdot\dot{x} + V(x)$$

we conclude that $E > V(x)$. This restricts the range of where the particle can possibly be. The points where $E = V(x)$ and $\dot{x} = \mathbf{0}$ are called *turning points*, since the particle typically 'bounces off' the potential barrier at these points. Depending on whether there are potential barriers on all sides or just a few, the motion of the particle may be *bounded* or *unbounded*.

A special case occurs when $E = V(x)$ and $V'(x) = 0$. These are called *equilibrium points* – the particle can sit indefinitely. At such a point, we have

$$\begin{aligned} m\ddot{x} &= -V'(x) = 0 \\ E &= \frac{1}{2}m\dot{x}^2 + V(x) \\ \implies &\boxed{\dot{x} = \ddot{x} = 0} \end{aligned}$$

Furthermore, motion near an equilibrium point is especially simple. Let x_0 be an equilibrium point, and Taylor expand:

$$V(x) \approx V(x_0) + \underbrace{(x - x_0)V'(x_0)}_{0} + \frac{1}{2}(x - x_0)^2 V''(x_0)$$

If $V''(x_0) > 0$, this is the potential for a harmonic oscillator. We end up with

$$\begin{aligned} m\ddot{x} &= -V'(x) = -(x - x_0)V''(x_0) \\ \implies x &= x_0 + A\cos(\omega t + \phi) \end{aligned}$$

so we have oscillations with frequency $\omega = (V''(x_0)/m)^{1/2}$.

If the amplitude A is small enough, then we can neglect higher-order terms in the Taylor expansion and the solution is self-consistent (we won't end up getting too far from the potential well). This is called a *stable equilibrium point*.

If instead $V''(x_0) < 0$, we have an unstable equilibrium point:

$$x = x_0 + Ae^{\gamma t} + Be^{-\gamma t}$$

where $\gamma = \sqrt{-V''(x_0)/m}$. If $A \neq 0$, then we will move far from the equilibrium point and so the Taylor expansion will break down.

The case where $A = 0$ corresponds to the particle having just enough energy to reach a maximum of the potential somewhere else.

If $V''(x_0) = 0$, we simply go to a higher order of the Taylor expansion.

§2.6 Dimensional analysis

Dimensional analysis is a way to get information about solutions to equations without actually solving them.

At a mathematical level, it is the ability to rescale variables to remove certain constants from equations.

Consider the equation for a pendulum,

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\sin\theta$$

and let us attempt to find the period of θ. First, we substitute $t = \sqrt{\frac{l}{g}}\tau$. We then have

$$\begin{aligned} \theta(t(\tau)) &= F(\tau) \\ \frac{d^2F}{d\tau^2} &= -\sin F \end{aligned}$$

Clearly now the solution must be some function $f(\tau)$ – it cannot possibly depend on g and l. We conclude that the period $\Delta\tau$ satisfies

$$\begin{aligned} F(\tau) &= F(\tau + \Delta\tau) \\ \implies F\left(\sqrt{\frac{g}{l}}t\right) &= F\left(\sqrt{\frac{g}{l}}t + \Delta\tau\right) \\ &= F\left(\sqrt{\frac{g}{l}}\left(t + \sqrt{\frac{l}{g}}\Delta\tau\right)\right) \end{aligned}$$

$$= \theta\left(t + \sqrt{\frac{l}{g}}\Delta\tau\right)$$

and so, without ever solving the eqation, we conclude that the period of a pendulum is proportional to the square root of its length.

In general the necessary rescaling may not be obvious. By endowing our constants and variables with so-called ‘dimensions’, we can perform this sort of analysis much more easily. Our basic dimensions are

- *length*, L;
- *time*, T;
- and *mass*, M.

Then we have results such as

$$\begin{aligned}
[\dot{x}] &= \mathrm{LT}^{-1} \\
[\ddot{x}] &= \mathrm{LT}^{-2} \\
[F] &= \mathrm{MLT}^{-2} \\
[E] &= \mathrm{ML}^{2}\mathrm{T}^{-2}
\end{aligned}$$

There may be other dimensions to keep track of, depending on the problem.

Theorem 2.8

For all physical laws, the dimensions of the left and right hand sides are equal.

Corollary 2.9

All arguments of non-trivial functions, i.e. those involving sums of different powers, must be dimensionless.

Example (Pendulum revisited)

We have $[g] = \mathrm{LT}^{-2}$, $[l] = \mathrm{L}$, $[m] = \mathrm{M}$, $[\theta_0] = 1$. We want the period T to have $[T] = \mathrm{T}$. So we let

$$\begin{aligned}
T &= f(\theta_0)g^A l^B m^C \\
\implies [T] &= [g]^A[l]^B[m]^C \\
&= \mathrm{L}^A\mathrm{T}^{-2A}\mathrm{L}^B\mathrm{M}^C
\end{aligned}$$

and we end up with the system

$$\begin{aligned}
0 &= C \\
0 &= A + B \\
1 &= 2A
\end{aligned}$$

which gives $T = f(\theta_0)\sqrt{\frac{l}{g}}$, as before.

If A, B, C are not all fixed, we have a dimensionless ratio and it will be necessary to apply some more thought.

Lecture 5 – 31/01/26

§2.7 Friction

Friction is not a fundamental force of nature, but a force that arises from complicated interactions that we don't want to consider in full detail.

In particular, an object passing through a medium such as air or water experiences microscopic forces between it and the medium – these forces cause momentum to be lost into the medium.

In part because of this, friction

- does not conserve energy (of the object);
- is *irreversible* – energy is lost by the object and never gained. This is closely related to the second law. In particular, the sign of a friction force must be such that it slows the object down. This implies, among other things, that friction depends on velocity.

Let us consider two important cases:

1. *Linear drag*. We have $\mathbf{F} = -k_1\mathbf{v}$.
2. *Quadratic drag*. We have $\mathbf{F} = -k_2|\mathbf{v}|\mathbf{v}$.

where k_1 and k_2 are called *coefficients of friction*.

Quadratic drag is more intuitive since collisions per unit time and momentum lost in each collision are both proportional to the particle's velocity.

Furthermore, the number of collisions should be proportional to the density of the medium, as well as the cross-sectional area of the object, so the force should also be proportional to these. Now, we can provisionally check units:

$$[F] = \mathrm{MLT}^{-2}$$
$$[k_2 v^2] = \mathrm{ML}^{-3}\mathrm{L}^2\mathrm{L}^2\mathrm{T}^{-2}$$

and these are equal, suggesting we have probably accounted for the main factors influencing k_2.

What about linear drag? Linear drag depends on viscous effects – the fact than an object drags the medium along with it as it moves and, as we move closer to the object, the velocity of the medium approaches v. It can then be shown that the momentum loss here is in fact linear in the velocity.

Example (Stokes' law)
For a spherical object, we have linear drag, and the coefficient is

$$k_1 = 6\pi\eta L$$

where η is the viscosity.

In general, we see both linear and quadratic drag, and which one dominates depends on the ratios of certain quantities. For example, taking k_1 equal to the value for Stokes' law above, we have

$$\frac{F_{\text{quad}}}{F_{\text{lin}}} \sim \frac{\rho A v^2}{\eta L v} \sim \frac{\rho L v}{\eta} = R$$

and this R is called the *Reynolds number*.

We are also yet to come onto *dry friction*, in other words the friction between two rigid bodies.

§2.7.1 Terminal velocity

Consider a particle falling under gravity with quadratic drag. For the z-component, we have

$$m\frac{dv}{dt} = -mg + kv^2$$

When the velocity is constant, the left hand side is zero, so we have

$$\begin{aligned} mg &= kv^2 \\ \implies v_{\text{term}} &= \sqrt{\frac{mg}{k}} \end{aligned}$$

Unlike escape velocity, we still have a mass here, so heavier objects reach a larger terminal velocity.

We can solve this equation analytically, but we can ask simple questions like: on what timescale does the object reach its terminal velocity? Since this should only depend on mass, gravity and k, we can apply dimensional analysis:

$$\begin{aligned} t &\propto m^A g^B k^C \\ \implies \mathrm{T} &= \mathrm{M}^A \mathrm{L}^B \mathrm{T}^{-2B} \mathrm{M}^C \mathrm{L}^{-C} \\ \implies &\begin{cases} A = -\frac{1}{2} \\ B = -\frac{1}{2} \\ C = \frac{1}{2} \end{cases} \end{aligned}$$

and so

$$t_0 = \sqrt{\frac{m}{kg}}$$

Hence, we know that v as a function of time takes the form

$$v = v_{\text{term}} F\left(\frac{t}{t_0}\right)$$

and so t_0 is the timescale on which the object reaches its terminal velocity.

We should also note that we can have terminal velocities arise from movement in the full 3 dimensions, giving the vector differential equation

$$m\frac{d\mathbf{v}}{dt} = m\mathbf{g} - k|\mathbf{v}|\mathbf{v}$$

§2.7.2 Damping

Friction also arises as a damping force in motion around an equilibrium point. For small oscillations, the linear drag is typically more important, giving the equation (in one dimension)

$$\ddot{x} = -\omega_0^2 x - 2\alpha\dot{x}$$

We have the solution

$$x = e^{-\alpha t}\left(A_+ e^{i\Omega t} + A_- e^{-i\Omega t}\right)$$

where $\Omega = \mathrm{Re}\left(\sqrt{\omega_0^2 - \alpha^2}\right)$. We have three cases:

- $\omega_0^2 > \alpha^2$: *Underdamping*. Here we have decaying oscillations
- $\omega_0^2 < \alpha^2$: *Underdamping*. Here we have exponential decay.
- $\omega_0^2 = \alpha^2$: *Critical damping*. Here we have a repeated root and instead get $x = (A + Bt)e^{-\alpha t}$.

> **Remark.** x and $\ddot{x}$ are invariant under *time reversal*, $t \to -t$. But $\dot{x}$ is not. As far as we know, all laws of nature are invariant under CPT, or charge-parity-time, symmetry, i.e. taking $q \to -q$, $\mathbf{x} \to -\mathbf{x}$ and $t \to -t$. But friction forces are odd under time reversal (and lack charge to compensate), so they are not fundamental.

§3 Central forces

An important class of potentials (gravity, electrostatics) only depend on the distance to the origin (or some other point) – we have

$$V(\mathbf{x}) = V(|\mathbf{x}|) = V(r)$$

Proposition 3.1

For such a potential, the corresponding force points towards or away from the origin.

Proof. We have

$$\begin{aligned}\mathbf{F} = -\nabla V &= -\frac{dV}{dr}\nabla r \\ &= -\frac{dV}{dr}\hat{\mathbf{x}}\end{aligned}$$

which is what we wanted. ∎

§3.1 Conservation of angular momentum

The most important fact about central forces is that they conserve angular momentum.

Definition 3.2 (Angular momentum)**.** The *angular momentum* of an object with position vector $\mathbf{x}$ is given by

$$\mathbf{L} = m\mathbf{x} \times \dot{\mathbf{x}} = \mathbf{x} \times \mathbf{p}$$

We can see that $\mathbf{L}$ is orthogonal to both position and velocity/momentum. It is important to note that $\mathbf{L}$ is defined relative to a particular origin – here we are taking the origin at $\mathbf{x} = \mathbf{0}$, the same point as the centre of the central potential.

For a general force $\mathbf{F}$, we have

$$\begin{aligned}\frac{d\mathbf{L}}{dt} &= m\frac{d}{dt}(\mathbf{x} \times \dot{\mathbf{x}}) \\ &= m(\dot{\mathbf{x}} \times \dot{\mathbf{x}} + \mathbf{x} \times \ddot{\mathbf{x}}) \\ &= \mathbf{x} \times \mathbf{F} = \mathbf{G}\end{aligned}$$

where $\mathbf{G}$ is defined to be the *torque*, the angular analogue of force.

Proposition 3.3

Central forces conserve angular momentum.

Proof. By proposition 3.1, we have $\mathbf{F} \parallel \mathbf{x}$ for such a force. Then $\frac{d\mathbf{L}}{dt} = \mathbf{x} \times \mathbf{F} = \mathbf{0}$. ■

Furthermore, because $\mathbf{L}$ is constant, and $\mathbf{x}$ and $\dot{\mathbf{x}}$ obey

$$\begin{aligned}\mathbf{L} \cdot \mathbf{x} &= 0 \\ \mathbf{L} \cdot \dot{\mathbf{x}} &= 0\end{aligned}$$

we can see that $\mathbf{x}$ and $\dot{\mathbf{x}}$ are constrained to a plane orthogonal to $\mathbf{L}$. So, for such a force, we can reduce to a two-dimensional case.

§3.2 Polar coordinates in the plane

Consider coordinates r and θ, with

$$\begin{aligned}x &= r\cos\theta \\ y &= r\sin\theta\end{aligned}$$

We would like to define some basis vectors $\hat{\mathbf{r}}$ and $\hat{\theta}$ in polar coordinates. In Cartesian coordinates, we have

$$\hat{\mathbf{r}} = \begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix}, \; \hat{\theta} = \begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}$$

and this is indeed an orthnormal basis. Of course, this basis will change as the particle moves, so care will be needed. In particular, we have

$$\frac{d\hat{\mathbf{r}}}{d\theta} = \hat{\theta}, \; \frac{d\hat{\theta}}{d\theta} = -\hat{\mathbf{r}}$$

Let us now attempt to write Newton's equations in polar coordinates. We have

$$\begin{aligned}
&\mathbf{x} = r\hat{\mathbf{r}} \\
\implies &\dot{\mathbf{x}} = \dot{r}\hat{\mathbf{r}} + r\dot{\hat{\mathbf{r}}} \\
&= \dot{r}\hat{\mathbf{r}} + r\frac{d\hat{\mathbf{r}}}{d\theta}\dot{\theta} \\
&= \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta} \\
\implies &\ddot{\mathbf{x}} = \ddot{r}\hat{\mathbf{r}} + 2\dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{\mathbf{r}} \\
&= \boxed{(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}}
\end{aligned}$$

Example 3.4 (Circular motion at constant angular speed)
We have

$$\begin{aligned}
\dot{r} &= 0,\ \dot{\theta} = \omega \\
\ddot{r} &= 0,\ \ddot{\theta} = 0
\end{aligned}$$

Then

$$\ddot{\mathbf{x}} = -r\omega^2\hat{\mathbf{r}}$$

– that is, circular motion requires centripetal force towards the origin.

To write down Newton's equations for a central force, we can equate radial and angular components. We have

$$\begin{aligned}
r\ddot{\theta} + 2\dot{r}\dot{\theta} &= 0 \\
m\left(\ddot{r} - r\dot{\theta}^2\right) &= -\frac{dV}{dr}
\end{aligned}$$

Focusing for now on the easier-looking $\hat{\theta}$ equation, we have

$$\begin{aligned}
&\frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}) = 0 \\
\implies &l = r^2\dot{\theta} = \text{const.}
\end{aligned}$$

In fact, this quantity l is (nearly) the magnitude of the angular momentum. We have

$$\begin{aligned}
&\mathbf{L} = m\mathbf{x} \times \dot{\mathbf{x}} \\
&= mr\hat{\mathbf{r}} \times (\dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta}) \\
&= mr^2\dot{\theta}\hat{\mathbf{r}} \times \hat{\theta} \\
\implies &|\mathbf{L}| = ml
\end{aligned}$$

but we often call l the angular momentum regardless.

Now, let us focus on the $\hat{\mathbf{r}}$ equation. Applying the definition of l, we have

$$\begin{aligned}
&m(\ddot{r} - r\frac{l^2}{r^4}) = -\frac{dV}{dr} \\
\implies &m\ddot{r} = -\frac{dV_{\text{eff}}}{dr}
\end{aligned}$$

where the *effective potential* is

$$V_{\text{eff}}(r) = V(r) + \frac{ml^2}{2r^2}$$

and now we've finally reduced to a one-dimensional problem.

§3.3 The effective potential

Consider a potential $V(r) = -\frac{k}{r}$, such as in gravity. Let us now plot V_{eff} against r – we have DIAGRAM. For small r, the $\frac{ml^2}{2r^2}$ term is most important, while for large r, the $-\frac{k}{r}$ dominates. This phenomenon is called the *centrifugal barrier* – conservation of angular momentum prevents the particle from getting too close to the origin. This is why orbits happen!

We can also see the effective potential by looking at the conserved energy. We have

$$\begin{aligned} E &= \frac{1}{2}m\dot{\mathbf{x}}\cdot\dot{\mathbf{x}} + V(r) \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + V(r) \\ &= \frac{1}{2}m\dot{r}^2 + V(r) + \frac{ml^2}{2r^2} \\ &= \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r) \end{aligned}$$

and here it's clear that the centrifugal barrier is just a product of the angular kinetic energy.

DIAGRAM: horizontal energy lines on the potential graph correspond to hyperbolic, parabolic, elliptical and circular orbits. Closest point for elliptical periapsis/perihelion, furthest apoapsis/aphelion.

Let us quickly consider a world in which the potential takes the form $-kr^{-n}$, $n > 2$. Then

DIAGRAM: potential graph, inverted.

Now we don't have any stable orbits! In fact, gravity in d spatial dimensions has $V \propto \frac{1}{r^{d-2}}$ – so circular orbits are only stable in dimensions $d < 4$.

Let us now solve the equations of motion.

§3.4 The orbit equation

The cleanest solution starts with a clever change of variables, to $u = \frac{1}{r}$. We now want to find the orbit equation for $u(\theta)$ (instead of $r(t)$). We have

$$\begin{aligned} \frac{dr}{dt} &= \frac{dr}{d\theta}\dot{\theta} = \frac{dr}{d\theta}\frac{l}{r^2} = -l\frac{du}{d\theta} \\ \frac{d^2r}{dt^2} &= \frac{d}{dt}\left(-l\frac{du}{d\theta}\right) = -l\frac{d^2u}{d\theta^2}\dot{\theta} = -l^2u^2\frac{d^2u}{d\theta^2} \end{aligned}$$

We now have

$$\begin{aligned} m\ddot{r} - \frac{ml^2}{r^3} &= F(r) \\ \implies -l^2mu^2\frac{d^2u}{d\theta^2} - mu^3l^2 &= F\left(\frac{1}{u}\right) \end{aligned}$$

$$\implies \frac{d^2u}{d\theta^2} + u = -\frac{1}{ml^2u^2}F\left(\frac{1}{u}\right)$$

Of course, in our case, we have $V = -\frac{km}{r}$ (where $k = GM$). This gives

$$\begin{aligned} \frac{d^2u}{d\theta^2} + u &= \frac{k}{l^2} \\ \implies u &= A\cos(\theta - \theta_0) + \frac{k}{l^2} \end{aligned}$$

We can see that u is largest at $\theta = \theta_0$, so r is smallest there – this is the periapsis. We will choose our axes such that $\theta_0 = 0$. We obtain

$$\boxed{r = \frac{r_0}{e\cos\theta + 1}}$$

where $r_0 = \frac{l^2}{k}$ is a fixed constant, and e is a constant of integration. This is a conic section!

The constant of integration e is called the *eccentricity*. We have that

- $e = 0$ is a *circular orbit*;
- $0 < e < 1$ is a *elliptical orbit*;
- $e = 1$ is a *parabolic* orbit;
- $e > 1$ is a *hyperbolic* orbit.

Let us consider the elliptical case. We have

$$\begin{aligned} r_0 - re\cos\theta &= r^2 \\ \implies (r_0 - ex)^2 &= x^2 + y^2 \\ \implies \frac{(x - x_c)^2}{a^2} + \frac{y^2}{b^2} &= 1 \end{aligned}$$

where $r = 0$ is the *focus*, not the centre. The semi-axes a and b are given in terms of e and r_0, as is x_c.

Planets in the solar system have small e. The most elliptical orbit, Mercury, has $e \approx 0.2$. Halley's comet, on the other hand, has $e \approx 0.97$.

Let us consider the parabolic case. We have

$$\begin{aligned} r_0^2 - 2r_0x + x^2 &= x^2 + y^2 \\ \implies r_0^2 - 2r_0x &= y^2 \end{aligned}$$

In the hyperbolic case, we see that $r \to \infty$ at $\cos\theta = -\frac{1}{e}$ (in particular, $\theta > \pi/2$, so with our convention of $\theta_0 = 0$, it goes to infinity on the left).

One important result allows us to evaluate the energy of a particular solution. We have

$$\begin{aligned} E &= \frac{1}{2}m\dot{r}^2 + \frac{ml^2}{2r^2} - \frac{km}{r} \\ &= \cdots \\ &= \frac{mk^2}{2l^2}(e^2 - 1) \end{aligned}$$

§3.5 Kepler's laws

Historically, these laws were derived before the orbit equation was ever solved. Kepler derived them empirically, while Newton justified them mathematically.

Proposition 3.5 (Kepler's first law)
Each planet moves in an ellipse, with the sun at one focus.

Proposition 3.6 (Kepler's second law)
The line between a planet and the sun sweeps out equal areas in equal times.
DIAGRAM

Proof. We have

$$\begin{aligned} \delta A &= \frac{1}{2} r^2 \delta\theta \\ \implies \dot{A} &= \frac{1}{2} r^2 \dot{\theta} \\ &= \frac{l}{2} \end{aligned}$$

which is constant by conservation of angular momentum. ■

Lecture n

Proposition 3.7 (Kepler's third law)
The orbital period T is proportional to the radius $R^{3/2}$.

Dimensional analysis. It's natural to first show this using dimensional analysis. The only parameter in Newton's equation is $k = GM$ (since m cancels). Hence, the period is

$$\begin{aligned} T &= cR^A k^B \\ \implies \mathrm{L} &= \mathrm{L}^A \mathrm{L}^{3B} \mathrm{T}^{-2B} \end{aligned}$$

which gives $B = -\frac{1}{2}$, $A = \frac{3}{2}$, giving $T = c\frac{R^{3/2}}{k^{1/2}}$ as required.

We should remark that, since there is no canonical 'radius' for an ellipse, this c depends on whether we pick the semi-major axis, semi-minor axis or anything in between. ■

Direct. It is in fact true that K3 only holds for inverse-square law orbits, which is a subtlety we would not notice doing dimensional analysis. So let us derive the law directly. Recall that $\frac{dA}{dt} = \frac{l}{2}$. Then

$$\begin{aligned} T = \int_0^T dt &= \int_0^A \frac{2}{l}\, dA \\ &= \frac{2\pi ab}{l} \end{aligned}$$

$$= \frac{2\pi}{l} \cdot \frac{r_0^2}{(1-e^2)^{3/2}}$$

Let us recall also that $l^2 = kr_0$, so we have

$$\begin{aligned} T &= \frac{2\pi}{\sqrt{GM}} \left(\frac{r_0}{1-e^2} \right)^{3/2} \\ &= \frac{2\pi}{\sqrt{GM}} R_{\text{avg}}^{3/2} \end{aligned}$$

because

$$\frac{r_0}{1-e^2} = \frac{1}{2} \left(\frac{r_0}{1+e} + \frac{r_0}{1-e} \right)$$

which are the minimum and maximum radii respectively. This concludes the proof. ∎

§3.6 Repulsive potentials and scattering

Given a central potential $V(r)$ satisfying $V \to 0$ as $r \to \infty$, one can perform *scattering experiments* – shoot a particle in from large r and see how it flies out again. We have an important new concept.

Definition 3.8 (Impact parameter)**.** The *impact parameter* for a scattering experiment, b, is the closest the particle would have come to the target in the absence of forces.

Proposition 3.9

The impact parameter b satisfies

$$l = bv$$

where l is the (specific) angular momentum, and v is the particle's initial velocity.

Proof. If we imagine the particle proceeding along the path without forces, it will have velocity v at the point of closest approach, in particular b. At this point, the vector $\mathbf{v}$ is orthogonal to the radial vector from the central object to the particle, so $|\mathbf{v} \times \mathbf{r}| = |\mathbf{v}||\mathbf{r}| = vb$. Since angular momentum is conserved, this must hold for the particle anywhere along its path, even when forces are applied. ∎

Let us consider the Rutherford scattering experiment, which showed the existence of the nucleus. For two positively charged particles, we have a potential of

$$V = \frac{\kappa}{r}$$

where κ is $\frac{qQ}{4\pi\varepsilon_0}$ (this is just Coulomb's law).

We may reuse results from the Kepler problem, replacing all instances of $-km$ with κ. In particular, the orbits are still

$$r = \frac{r_0}{\tilde{e}\cos\theta - 1}$$

with $r_0 = \frac{l^2 m}{\kappa}$ (note we've changed some signs but it is in fact the same equation). We want to find the scattering angle ϕ between the particle's incoming and outgoing paths.

First, note the incoming and outgoing impact parameters are equal, since angular momentum is conserved and, at a given distance, the speed is the same by symmetry or the conservation of energy. Looking at the orbit equation, we can clearly see that r goes to infinity when the denominator is zero, i.e. $\tilde{e}\cos\theta = 1$. Let us consider energies. We have

$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{\kappa^2}{2l^2m}\left(\tilde{e}^2 - 1\right)\\ &= \frac{\kappa^2}{2m(bv)^2}\left(\frac{1}{\cos^2\theta} - 1\right)\\ &= \frac{\kappa^2}{2mb^2v^2}\tan^2\theta\end{aligned}$$

Also, we can see (DIAGRAM!!!!) that the angle of deviation of the particle's path, ϕ, satisfies $\phi + 2\theta = \pi$, so

$$\tan\theta = \tan\left(\frac{\pi - \phi}{2}\right) = \frac{1}{\tan\frac{\phi}{2}}$$

Hence

$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{\kappa^2}{2mb^2v^2}\cdot\frac{1}{\tan^2\frac{\phi}{2}}\\ \implies \phi &= 2\arctan\left(\frac{\kappa}{bmv^2}\right)\end{aligned}$$

We see that a small b leads to a large scattering angle, which allows scattering experiments to probe very small distances.

§4 Systems of particles

Let us consider now systems of N particles, labelled with $i = 1, \ldots, n$. Each particle has position $\mathbf{x}_i$, mass m_i and momentum $\mathbf{p}_i = m_i\dot{\mathbf{x}}_i$. And of course each particle obeys Newton's law.

The force $\mathbf{F}_i$ on the i^{th} particle can be either *external* or due to the other particles:

$$\mathbf{F}_i = \mathbf{F}_i^{\text{ext}} + \sum_{i\neq j}\mathbf{F}_{ij}$$

where $\mathbf{F}_{ij}$ is the force on the i^{th} particle due to the j^{th} particle.

We have the following important result:

Theorem 4.1 (Newton's third law)
For two particles with labels i, j,

$$\mathbf{F}_{ij} = -\mathbf{F}_{ji}$$

– or, in words, 'every action has an equal and opposite reaction.'

Lecture $n+1$

Let us consider some consequence of Newton's third law for systems of particles.

§4.1 Centre of mass

The total mass is $M = \sum_{i=1}^{N} M_i$. We define the *centre of mass*

$$\mathbf{R} = \frac{1}{M}\sum_{i=1}^{N} M_i \mathbf{x}_i$$

so the total momentum is

$$\begin{aligned}\mathbf{P} &= \sum_{i=1}^{N} \mathbf{p}_i \\ &= \sum_i m_i \dot{\mathbf{x}}_i \\ &= m\dot{\mathbf{R}}\end{aligned}$$

so the centre of mass behaves like a single particle of mass M. Furthermore,

$$\begin{aligned}\dot{\mathbf{P}} &= \sum_i \dot{\mathbf{p}}_i \\ &= \sum_i (\mathbf{F}_i^{\text{ext}} + \sum_{j\neq i} \mathbf{F}_{ij}) \\ &= \sum_i \mathbf{F}_i^{\text{ext}} + \sum_{i<j} \mathbf{F}_{ij} + \mathbf{F}_{ji} \\ &= \mathbf{F}\text{, the total external force}\end{aligned}$$

where the second term in the penultimate line is zero by Newton's third law. We conclude that $M\ddot{\mathbf{R}} = \mathbf{F}$, the total external force – we are allowed to view composite objects like point particles. In particular, $\mathbf{F} = \mathbf{0} \implies \dot{\mathbf{p}} = \mathbf{0}$ – total momentum is conserved in the absence of external forces.

We would like to show the same result for total *angular* momentum. Let us work relative to an arbitrary fixed point $\mathbf{a}$: we have

$$\begin{aligned}\mathbf{L} &= \sum_i (\mathbf{x}_i - \mathbf{a}) \times \mathbf{p}_i \\ \implies \dot{\mathbf{L}} &= \sum_i (\mathbf{x}_i - \mathbf{a}) \times \dot{\mathbf{p}}_i \\ &= \sum_i (\mathbf{x}_i - \mathbf{a}) \times \left(\mathbf{F}_i^{\text{ext}} + \sum_{j\neq i} \mathbf{F}_{ij}\right) \\ &= \mathbf{G} + \sum_{i<j} \left((\mathbf{x}_i - \mathbf{a}) \times \mathbf{F}_{ij} - (\mathbf{x}_j - \mathbf{a}) \times \mathbf{F}_{ij}\right) \\ &= \mathbf{G} + \sum_{i<j} (\mathbf{x}_i - \mathbf{x}_j)\, \mathbf{F}_{ij} \qquad (\dagger)\end{aligned}$$

Here the total torque

$$\mathbf{G} = \sum_i (\mathbf{x}_i - \mathbf{a}) \times \mathbf{F}_i^{\text{ext}}$$

but the vanishing of the second term in $(\dagger)$ is not a given.

Let us consider the special case where the interaction forces come from a potential that depends on the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$, that is

$$\mathbf{F}_{ij} = -\nabla_i V(|\mathbf{x}_i - \mathbf{x}_j|)$$

where we are taking the gradient with respect to $\mathbf{x}_i$ This is

$$\mathbf{F}_{ij} = -V'(|\mathbf{x}_i - \mathbf{x}_j|)\frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|}$$

from which it clearly follows that $(\mathbf{x}_i - \mathbf{x}_j) \times \mathbf{F}_{ij} = \mathbf{0}$. In this case, $\dot{\mathbf{L}} = \mathbf{G}$.

In fact, there are tools that allow us to show that the second term in (†) will vanish for all known forces, but we cannot do this right now. In general, we conclude $\dot{\mathbf{L}} = \mathbf{G}$ – an object cannot spin itself.

It's natural to take $\mathbf{a}$, our 'centre', to be the centre of mass. However, in general, the centre of mass will change as a function of time, which will require us to redo the derivation above. But all will be well.

The total *energy* also splits up nicely: we have

$$\mathbf{x}_i = \mathbf{R} + \mathbf{y}_i$$

where $\mathbf{y}_i$ is the position of the i^{th} particle relative to the centre of mass. But, since

$$M\mathbf{R} = \sum_i m_i \mathbf{x}_i \tag{1}$$

we conclude that $\sum_i m_i \mathbf{y}_i = \mathbf{0}$ for all time. Then

$$\begin{aligned} T &= \sum_i \frac{1}{2} m_i \dot{\mathbf{x}}_i \cdot \dot{\mathbf{x}}_i && (2) \\ &= \sum_i \frac{1}{2} m_i \left(\dot{\mathbf{R}}^2 + \dot{\mathbf{y}}_i^2 + 2\dot{\mathbf{R}}\dot{\mathbf{y}}_i\right) && (3) \\ &= \frac{1}{2} M\dot{\mathbf{R}}^2 + \sum_i \frac{1}{2} m_i \dot{\mathbf{y}}_i^2 && (4) \end{aligned}$$

so we also have notions of the internal kinetic energy and the kinetic energy which can be in some sense attributed to the centre of mass.

Let us now consider potential energy, and work as usual under the assumption that all forces are conservative. We have

$$\begin{aligned} \mathbf{F}_i^{\text{ext}} &= -\nabla_i V_i(\mathbf{x}_i) \\ \mathbf{F}_{ij} &= -\nabla_i V_{ij}(|\mathbf{x}_i - \mathbf{x}_j|) \end{aligned}$$

where $V_{ij} = V_{ji}$ to respect N3. We now have a conserved energy.

Proposition 4.2

The total energy

$$E = T + \sum_i V_i(\mathbf{x}_i) + \sum_{i<j} V_{ij}(|\mathbf{x}_i - \mathbf{x}_j|)$$

is conserved. *Note:* each potential term only appears once!

Proof. We have

$$\begin{aligned}
\frac{dE}{dt} &= \sum_i m_i \dot{\mathbf{x}}_i \cdot \ddot{\mathbf{x}}_i + \sum_i \nabla_i V_i \cdot \dot{\mathbf{x}}_i + \sum_{i<j} \left(\nabla_i V_{ij} \cdot \dot{\mathbf{x}}_i + \nabla_j V_{ij} \cdot \dot{\mathbf{x}}_j \right) \\
&= \sum_i m_i \dot{\mathbf{x}}_i \cdot \ddot{\mathbf{x}}_i - \sum_i \mathbf{F}_i^{\text{ext}} \cdot \dot{\mathbf{x}}_i - \sum_{i<j} \left(\mathbf{F}_{ij} \cdot \dot{\mathbf{x}}_i + \mathbf{F}_{ji} \cdot \dot{\mathbf{x}}_j \right) \\
&= \sum_i \dot{\mathbf{x}}_i \cdot \left(m \ddot{\mathbf{x}}_i - \mathbf{F}_i^{\text{ext}} - \sum_{i \neq j} \mathbf{F}_{ij} \right) \\
&= 0
\end{aligned}$$

by Newton's second law. ∎